

Dumindu Kumarapeli

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Data science undergraduate building research prototypes that run end to end — agentic and retrieval-augmented LLM systems, applied ML, and computer vision.

Education

NSBM Green University, Sri Lanka

BSc (Hons) Data Science, 2025 – expected December 2028

- GPA 3.85

Research

D. Kumarapeli and H. Weerasinghe, “Privacy-preserving automated exploratory data analysis using deterministic summarization and local large language models,” *Proc. NSBM Student Research Symposium on Innovation Technology (NSRSIT)*, 2026, pp. 146–150. **Best Overall Research**, session award. Written in the second year of the degree.

Composite clinical proxy for durable clinical benefit in oncology — modelling durable clinical benefit from real-world clinicogenomic data. *[under review]*

ML-enhanced SEIR pipeline for dengue forecasting, Sri Lanka — hybrid mechanistic and learned model for short-horizon prediction and spatial risk mapping. *[in progress]*

Recovering neuromodulatory profiles from RL-agent behaviour — joint identifiability and degeneracy of parameters inferred from behaviour alone. *[in progress]*

Projects

OWL — Datacenter AI Operations Platform

Python, FastAPI, PyTorch, scikit-learn, ChromaDB, Gemini, MCP, React 19 github.com/Dumee-25/OWL-Datacenter_agent

- Built an agentic RAG assistant over ChromaDB using hybrid vector and keyword retrieval with recency re-ranking, served through REST, WebSocket, and a nine-tool MCP server.
- Paired an Isolation Forest (point) and LSTM autoencoder (sequence) ensemble with a state-machine alert manager; alerts carry a confidence score and the system never actuates infrastructure.
- Wrote a simulator generating 4.7M rows of telemetry over 18 months with a temporal holdout split, and instrumented per-call latency and cost in the async FastAPI backend.

DataSense — Privacy-Preserving Automated EDA

FastAPI, Next.js 14, PostgreSQL, Polars, SciPy, scikit-learn, Ollama github.com/Dumee-25/DataSense

- Implemented a seven-step async pipeline that profiles a CSV, runs quality, correlation and outlier checks, flags subgroup reversal, and returns a confidence-scored model recommendation with up to 14 charts and a PDF report.
- Kept raw rows out of the language model by computing the statistics first and passing only the resulting facts to a pluggable backend (local Ollama by default, OpenAI or Groq optional), with a rule-based fallback when no model is available.
- Hardened the upload path with per-IP rate limiting, binary and formula-injection checks, and concurrency control.

Seshat — Local-First Research Memory

Python, FastAPI, React, Ollama, ChromaDB, SQLite, PyInstaller github.com/Dumee-25/Seshat

- Built a passive filesystem watcher that diffs notebooks at cell level and indexes scripts, outputs and papers into append-only event storage, so the raw record survives changes to the layers above it.
- Journals each session with a local qwen3:8b model and a confidence-scored inferred intent the researcher can correct inline; papers embed with bge-small-en-v1.5 on CPU, so nothing leaves the machine.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, R, SQL

ML / data: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, pandas, Polars, NumPy, SciPy

LLM / RAG: Ollama, Gemini, ChromaDB, Model Context Protocol (MCP), retrieval-augmented generation

Computer vision: YOLOv8, OpenCV, ByteTrack

Parallel computing: CUDA, OpenMP, MPI

Backend: FastAPI, Spring Boot, PostgreSQL, MySQL, SQLite, Docker, Git, Linux

Frontend: React, Next.js, Node.js, Streamlit, Plotly, Power BI

Currently working through: ROS 2, C++, sequence models (RNN/LSTM)

Spoken: English (Fluent), Sinhala (Native)

Awards and Leadership

Best Overall Research, session award — NSRSIT 2026, NSBM Green University. *[2026]*

Data Storm 7.0 — Final Round, Data Drifters, Top 10. *[2026]*